

Derivation of the Keller–Segel Equation from HJB Framework

1 Background and Motivation

1.1 What is the Keller–Segel model?

The **Keller–Segel (KS) model** is a system of partial differential equations (PDEs) that describes *chemotaxis*: the tendency of biological cells (e.g. bacteria or slime-mould amoebae) to move *up* (or down) a gradient of a chemical signal. A classic example is *E. coli* swimming toward regions of higher nutrient concentration.

The standard KS system couples two quantities:

$$\begin{aligned}\rho(x, t) &= \text{cell (population) density at position } x \text{ and time } t, \\ Z(x, t) &= \text{concentration of the chemoattractant (chemical cue)}.\end{aligned}$$

The goal of these notes is to *derive* the KS equations from first principles using the language of **Reinforcement Learning (RL)** and the **Hamilton–Jacobi–Bellman (HJB)** framework.

1.2 Why Reinforcement Learning?

In RL, an *agent* navigates an environment by choosing actions to maximise cumulative reward. Here, each cell is the agent, space is the environment, and the reward is the local nutrient concentration. The cell must balance two drives:

1. Moving toward high-reward regions (*exploitation*).
2. Not deviating too far from its natural (unbiased) random walk (*exploration* / control cost).

This trade-off is captured precisely by *entropy-regularised* RL.

Intuition

Think of a bacterium as a tiny robot that wants to find food. It cannot plan far ahead, and it pays a price (in energy) every time it deviates from its default tumble-and-run motion. The RL framework quantifies this trade-off mathematically, and the Keller–Segel equations emerge as its continuous-space limit.

2 The Reinforcement Learning Objective

2.1 Setup

We work on a discrete state space S (a lattice of positions). At each discrete time step t , a cell is at state $s_t \in S$. The dynamics are:

- **Unbiased (intrinsic) transition:** $p(s'|s)$ is the probability of moving from s to s' under a symmetric random walk (no chemotaxis).
- **Controlled (policy) transition:** $\pi(s'|s)$ is the probability the cell *actually* chooses to move to s' , which may differ from p .

2.2 Entropy-regularised objective

The cell seeks to maximise the functional

$$J[\pi] := \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t \left(r(s_t) - \frac{1}{\beta} \log \frac{\pi(s_{t+1}|s_t)}{p(s_{t+1}|s_t)} \right) \right], \quad (1)$$

where:

- $r(s)$ is the **reward** at state s (e.g. local nutrient concentration).
- $\gamma \in (0, 1]$ is the **discount factor**: future rewards are worth γ^t of an immediate reward at the same level.
- $\beta > 0$ is the **inverse temperature** (or inverse control cost): large β means the cell is willing to deviate strongly from p ; small β keeps it close to the unbiased walk.
- $\log \frac{\pi(s'|s)}{p(s'|s)}$ is the **KL divergence term** per step. It penalises the cell for deviating from its natural motion.

Definition

The **Kullback–Leibler (KL) divergence** between distributions π and p over S measures how different π is from p :

$$D_{\text{KL}}(\pi \| p) = \sum_{s'} \pi(s'|s) \log \frac{\pi(s'|s)}{p(s'|s)} \geq 0.$$

It is zero only when $\pi = p$. In eq. (1), the expectation of $\log \frac{\pi(s_{t+1}|s_t)}{p(s_{t+1}|s_t)}$ under π is exactly this divergence (evaluated one step at a time), so the cell pays a cost proportional to how much it deviates from the unbiased walk.

3 The Optimal Policy and the HJB Equation

3.1 The Bellman optimality equation

The key quantity in RL is the **value function**

$$V^\pi(s) := \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t \left(r(s_t) - \frac{1}{\beta} \log \frac{\pi(s_{t+1}|s_t)}{p(s_{t+1}|s_t)} \right) \mid s_0 = s \right],$$

which is the expected total discounted return starting from state s under policy π . The *optimal* value function $V^*(s) = \max_{\pi} V^{\pi}(s)$ satisfies the **Bellman optimality equation**.

Derivation of the Bellman equation (step by step)

We start from the definition of $V^*(s)$:

$$V^*(s) = \max_{\pi} \left\{ r(s) + \gamma \sum_{s'} \pi(s'|s) \left[V^*(s') - \frac{1}{\beta} \log \frac{\pi(s'|s)}{p(s'|s)} \right] \right\}.$$

Define the one-step objective for a given π :

$$F[\pi] = \sum_{s'} \pi(s'|s) \left[\gamma V^*(s') - \frac{1}{\beta} \log \frac{\pi(s'|s)}{p(s'|s)} \right].$$

We want to maximise F over all probability distributions $\pi(\cdot|s)$ (subject to $\sum_{s'} \pi(s'|s) = 1$).

Lagrange multiplier optimisation. Introduce a multiplier λ for the normalisation constraint:

$$\frac{\partial}{\partial \pi(s'|s)} \left[\pi(s'|s) \left(\gamma V^*(s') - \frac{1}{\beta} \log \frac{\pi(s'|s)}{p(s'|s)} \right) - \lambda \pi(s'|s) \right] = 0.$$

Differentiate term by term. Recall $\frac{d}{dq}[q \log q] = 1 + \log q$:

$$\begin{aligned} & \gamma V^*(s') - \frac{1}{\beta} \left(\log \frac{\pi(s'|s)}{p(s'|s)} + 1 \right) - \lambda = 0 \\ \implies & \log \frac{\pi(s'|s)}{p(s'|s)} = \beta \gamma V^*(s') - \beta \lambda - 1. \end{aligned}$$

Exponentiating:

$$\pi^*(s'|s) = p(s'|s) e^{\beta \gamma V^*(s') - \beta \lambda - 1}.$$

Normalising (summing over s' and requiring $\sum_{s'} \pi^*(s'|s) = 1$):

$$e^{\beta \lambda + 1} = \sum_{s''} p(s''|s) e^{\beta \gamma V^*(s'')},$$

which gives:

$$\boxed{\pi^*(s'|s) = \frac{p(s'|s) e^{\beta \gamma V^*(s')}}{\sum_{s''} p(s''|s) e^{\beta \gamma V^*(s'')}}.} \quad (2)$$

Substituting back to get the HJB/Bellman equation. Insert π^* into $V^*(s) = r(s) + F[\pi^*]$. The log-ratio for π^* is:

$$\log \frac{\pi^*(s'|s)}{p(s'|s)} = \beta \gamma V^*(s') - \log \sum_{s''} p(s''|s) e^{\beta \gamma V^*(s'')}.$$

Therefore:

$$\begin{aligned}
F[\pi^*] &= \sum_{s'} \pi^*(s'|s) \left[\gamma V^*(s') - \frac{1}{\beta} \left(\beta \gamma V^*(s') - \log \sum_{s''} p(s''|s) e^{\beta \gamma V^*(s'')} \right) \right] \\
&= \sum_{s'} \pi^*(s'|s) \cdot \frac{1}{\beta} \log \sum_{s''} p(s''|s) e^{\beta \gamma V^*(s'')} \\
&= \frac{1}{\beta} \log \sum_{s''} p(s''|s) e^{\beta \gamma V^*(s'')}.
\end{aligned}$$

Hence:

$$\boxed{V^*(s) = r(s) + \frac{1}{\beta} \log \sum_{s' \in S} p(s'|s) e^{\beta \gamma V^*(s')}} \quad (3)$$

This is the **soft Bellman equation** (or free-energy Bellman equation).

Intuition

Compare with the standard (hard) Bellman equation $V^*(s) = r(s) + \gamma \max_{s'} V^*(s')$. The soft version replaces the hard max with a *log-sum-exp*, which is a smooth approximation to the maximum. As $\beta \rightarrow \infty$ (no control cost), $\log \sum e^{\beta x} \rightarrow \max x$, recovering the standard equation.

4 Biological Mapping: Value Function to Chemical Cue

4.1 The Weber–Fechner logarithmic encoding

Cells cannot directly measure $V^*(s)$; instead they respond to chemical concentrations. We assume the cell encodes the value function logarithmically via a cue concentration $Z_t(s)$:

$$V_t(s) = \frac{1}{\beta} \log Z_t(s) \quad \Longleftrightarrow \quad Z_t(s) = e^{\beta V_t(s)}. \quad (4)$$

Note

This is the **Weber–Fechner law**: biological perception of stimulus intensity is proportional to the *logarithm* of the stimulus. It appears in sensory neuroscience (loudness, brightness) and here naturally in the RL-to-biology bridge.

4.2 Target equilibrium for the cue

Substitute (4) into the Bellman equation (3):

$$\frac{1}{\beta} \log Z_t(s) = r(s) + \frac{1}{\beta} \log \sum_{s'} p(s'|s) e^{\beta \gamma \cdot \frac{1}{\beta} \log Z_t(s')}.$$

Simplify the exponent: $\beta \gamma \cdot \frac{1}{\beta} \log Z = \gamma \log Z = \log Z^\gamma$.

$$\frac{1}{\beta} \log Z_t(s) = r(s) + \frac{1}{\beta} \log \sum_{s'} p(s'|s) Z_t(s')^\gamma.$$

Multiply through by β and exponentiate:

$$Z_t(s) = e^{\beta r(s)} \sum_{s'} p(s'|s) Z_t(s')^\gamma. \quad (5)$$

This is the *target* the cue concentration should satisfy at equilibrium.

4.3 Temporal Difference (TD) cue update rule

At any time t , the cue $Z_t(s)$ may not satisfy (5). Define the **Bellman error** (or TD error in cue space):

$$\delta_t(s) = e^{\beta r(s)} \sum_{s'} p(s'|s) Z_t(s')^\gamma - Z_t(s).$$

A gradient descent step to reduce this error gives the **cue update rule**:

$$Z_{t+\Delta t}(s) = Z_t(s) + \alpha \delta_t(s) = Z_t(s) + \alpha \left[e^{\beta r(s)} \sum_{s'} p(s'|s) Z_t(s')^\gamma - Z_t(s) \right], \quad (6)$$

where $\alpha \in (0, 1)$ is the **learning rate**.

Intuition

This is analogous to TD(0) in standard RL: each cell adjusts its local cue concentration toward the target prescribed by its current estimate of V^* . Over many iterations, $Z_t \rightarrow Z^*$ (the fixed point of (5)).

5 Transition to Continuous Space–Time: The Cue (Z) Equation

5.1 Setting up the Taylor expansion

We now take $\Delta x \rightarrow 0$, $\Delta t \rightarrow 0$ to pass to a PDE for $Z(x, t)$.

Consider a symmetric, unbiased random walk on a d -dimensional lattice with step size Δx and time step Δt . The transition probability is:

$$p(s'|s) = p(\Delta x), \quad \Delta x = s' - s,$$

with $p(\Delta x) = p(-\Delta x)$ (symmetry) and $\sum_{\Delta x} p(\Delta x) = 1$.

Let $f(s') = Z(s')^\gamma$. We want to evaluate

$$\sum_{s'} p(s'|s) f(s') = \sum_{\Delta x} p(\Delta x) f(s + \Delta x).$$

Expand $f(s + \Delta x)$ in a Taylor series around s :

$$f(s + \Delta x) = f(s) + (\Delta x \cdot \nabla) f(s) + \frac{1}{2} (\Delta x \cdot \nabla)^2 f(s) + O(\Delta x^3).$$

More explicitly in components (using Einstein summation):

$$f(s + \Delta x) = f(s) + (\Delta x)_i \partial_i f(s) + \frac{1}{2} (\Delta x)_i (\Delta x)_j \partial_i \partial_j f(s) + \dots$$

5.2 Using the symmetry of the random walk

Sum over Δx weighted by $p(\Delta x)$:

$$\begin{aligned} \sum_{\Delta x} p(\Delta x) f(s + \Delta x) &= f(s) \underbrace{\sum_{\Delta x} p(\Delta x)}_{=1} + \underbrace{\sum_{\Delta x} p(\Delta x) (\Delta x)_i \partial_i f(s)}_{=0 \text{ by symmetry}} \\ &\quad + \underbrace{\frac{1}{2} \sum_{\Delta x} p(\Delta x) (\Delta x)_i (\Delta x)_j \partial_i \partial_j f(s)}_{=\sigma^2 \delta_{ij}} + O(\Delta x^3), \end{aligned}$$

where $\sigma^2 = \sum_{\Delta x} p(\Delta x) (\Delta x)_1^2$ is the variance in one direction.

The first-order term vanishes because p is symmetric ($p(\Delta x) = p(-\Delta x)$ so each odd moment is zero).

The second-order term gives the Laplacian:

$$\sum_{\Delta x} p(\Delta x) (\Delta x)_i (\Delta x)_j = \sigma^2 \delta_{ij} \implies \frac{1}{2} \sigma^2 \delta_{ij} \partial_i \partial_j f = \frac{1}{2} \sigma^2 \nabla^2 f.$$

Defining the diffusion coefficient $D_Z = \frac{\sigma^2}{2\Delta t}$ (so that $\frac{1}{2} \sigma^2 = D_Z \Delta t$), we get:

$$\sum_{s'} p(s'|s) f(s') \approx f(s) + D_Z \Delta t \nabla^2 f(s) = Z(s)^\gamma + D_Z \Delta t \nabla^2 [Z(s)^\gamma]. \quad (7)$$

This is equation (7) in the original notes.

5.3 Linearising the reward factor

For small rewards $r(s)$ (or small β), expand the exponential:

$$e^{\beta r(s)} \approx 1 + \beta r(s) + O((\beta r)^2).$$

5.4 Substituting into the cue update rule

Insert (7) and the linearised exponential into (6):

$$\begin{aligned} Z_{t+\Delta t}(s) - Z_t(s) &= \alpha [(1 + \beta r)(Z^\gamma + D_Z \Delta t \nabla^2 Z^\gamma) - Z] \\ &= \alpha [Z^\gamma + D_Z \Delta t \nabla^2 Z^\gamma + \beta r Z^\gamma + \beta r D_Z \Delta t \nabla^2 Z^\gamma - Z]. \end{aligned}$$

Divide by Δt and take $\Delta t \rightarrow 0$. Writing $\alpha = \hat{\alpha} \Delta t$ (so that the learning rate is proportional to the time step, as is standard):

$$\frac{Z_{t+\Delta t} - Z_t}{\Delta t} = \hat{\alpha} [Z^\gamma - Z + D_Z \Delta t \nabla^2 Z^\gamma + \beta r Z^\gamma + O(\Delta t)].$$

5.5 The $\gamma \approx 1$ and dominant-term limit

Set $\gamma = 1$ (undiscounted, or near-undiscounted case). Then $Z^\gamma = Z$ and:

$$\frac{\partial Z}{\partial t} = \hat{\alpha} [Z - Z + D_Z \nabla^2 Z + \beta r Z - \dots].$$

The $Z - Z$ terms cancel. What remains from the expansion at order Δt^0 is:

- A diffusion term: $D_Z \nabla^2 Z$.
- A production term: $\beta r Z \rightarrow k_p \rho r(x)$, where we replace the reward r (which, physically, is a source term proportional to the local cell density ρ) and absorb $\beta \hat{\alpha}$ into the production rate k_p .
- A degradation term: $-k_d Z$ (coming from natural decay of the chemical cue; this arises at next order in the expansion or from a separate degradation term).

Collecting these, and identifying $D = D_Z$:

$$\boxed{\frac{\partial Z}{\partial t} = D \nabla^2 Z + k_p \rho r(x) - k_d Z.} \quad (8)$$

Note

Equation (8) is the standard *chemoattractant equation* in Keller–Segel models. It says the chemical cue diffuses (Laplacian term), is produced by cells at a rate k_p in proportion to cell density and local reward, and degrades spontaneously at rate k_d .

6 Derivation of the Cell Density Equation

6.1 Discrete master equation

The cell density $\rho(s, t)$ evolves by conservation of probability. Cells at s' at time t move to s with probability $\pi_t(s|s')$:

$$\rho(s, t + \Delta t) = \sum_{s'} \rho(s', t) \pi_t(s|s'). \quad (9)$$

6.2 Policy in terms of the cue

From (4) and (2):

$$\pi^*(s'|s) = \frac{p(s'|s) e^{\beta \gamma V^*(s')}}{\sum_{s''} p(s''|s) e^{\beta \gamma V^*(s'')}} = \frac{p(s'|s) Z(s')^\gamma}{\sum_{s''} p(s''|s) Z(s'')^\gamma}.$$

(We used $e^{\beta \gamma V^*} = e^{\beta \gamma \frac{1}{\beta} \log Z} = Z^\gamma$.)

In the master equation (9) we need $\pi_t(s|s')$ (transition *to* s *from* s'), i.e. with the roles of s and s' swapped:

$$\pi_t(s|s') = \frac{p(s|s') Z(s)^\gamma}{\sum_{s''} p(s''|s') Z(s'')^\gamma}.$$

6.3 Expanding the policy for small steps

Let $\Delta x = s - s'$ (displacement from s' to s). Expand $Z(s)^\gamma = Z(s' + \Delta x)^\gamma$ around s' :

$$Z(s' + \Delta x)^\gamma = Z(s')^\gamma + \nabla[Z^\gamma](s') \cdot \Delta x + O(\Delta x^2).$$

Factor out $Z(s')^\gamma$:

$$Z(s' + \Delta x)^\gamma = Z(s')^\gamma \left(1 + \frac{\nabla Z^\gamma}{Z^\gamma}(s') \cdot \Delta x + O(\Delta x^2) \right) = Z(s')^\gamma (1 + \gamma \nabla \log Z(s') \cdot \Delta x + O(\Delta x^2)),$$

using $\frac{\nabla Z^\gamma}{Z^\gamma} = \gamma \frac{\nabla Z}{Z} = \gamma \nabla \log Z$.

The denominator is a local average:

$$\sum_{s''} p(s''|s') Z(s'')^\gamma \approx Z(s')^\gamma \quad (\text{to leading order}).$$

(The correction is $O(\Delta x^2)$, as we showed in (7).)

Therefore, the policy linearises to:

$$\pi_t(s|s') \approx p(\Delta x)(1 + \gamma \nabla \log Z(s') \cdot \Delta x). \quad (10)$$

Note

Equation (10) says that the cell at s' is more likely to move in the direction of $\nabla \log Z$ (up the gradient of the log-cue = up the gradient of the value function). This is precisely chemotaxis!

6.4 Kramers–Moyal expansion of the master equation

We now pass to continuous space. Write the master equation as:

$$\rho(x, t + \Delta t) = \int \rho(x - \Delta x, t) \pi(x|x - \Delta x) d(\Delta x).$$

Expand $\rho(x - \Delta x, t)$ in a Taylor series around x :

$$\rho(x - \Delta x) = \rho(x) - (\Delta x \cdot \nabla) \rho(x) + \frac{1}{2} (\Delta x \cdot \nabla)^2 \rho(x) + O(\Delta x^3).$$

Similarly, expand $\pi(x|x - \Delta x) \approx p(\Delta x)(1 + \gamma \nabla \log Z(x) \cdot \Delta x)$ using (10) evaluated at $x - \Delta x \approx x$.

Substituting:

$$\rho(x, t + \Delta t) = \int \left[\rho - (\Delta x_i) \partial_i \rho + \frac{1}{2} (\Delta x_i) (\Delta x_j) \partial_i \partial_j \rho + \dots \right] p(\Delta x) [1 + \gamma (\partial_k \log Z)(\Delta x_k)] d(\Delta x).$$

Expand and collect terms by order in Δx :

Zeroth order: $O(1)$.

$$\underbrace{\rho \int p(\Delta x) d(\Delta x)}_{=1} = \rho.$$

First order: $O(\Delta x)$. Terms involving one power of Δx_i :

$$- \partial_i \rho \underbrace{\int (\Delta x_i) p(\Delta x) d(\Delta x)}_{=0} + \rho \gamma (\partial_k \log Z) \underbrace{\int (\Delta x_k) p(\Delta x) d(\Delta x)}_{=0} = 0.$$

Both integrals vanish by the symmetry of p .

Second order: $O(\Delta x^2)$. Terms involving two powers of Δx :

$$\frac{1}{2} \partial_i \partial_j \rho \underbrace{\int (\Delta x_i)(\Delta x_j) p d(\Delta x)}_{\sigma^2 \delta_{ij}} - (\partial_i \rho) \gamma (\partial_k \log Z) \underbrace{\int (\Delta x_i)(\Delta x_k) p d(\Delta x)}_{\sigma^2 \delta_{ik}}.$$

This gives (contracting indices):

$$\frac{\sigma^2}{2} \nabla^2 \rho - \sigma^2 \gamma (\nabla \rho) \cdot (\nabla \log Z).$$

Recalling $D_\rho = \frac{\sigma^2}{2\Delta t}$ (diffusion coefficient for the cells):

Putting it together.

$$\frac{\rho(x, t + \Delta t) - \rho(x, t)}{\Delta t} = D_\rho \nabla^2 \rho - 2D_\rho \gamma \nabla \rho \cdot \nabla \log Z.$$

We now rewrite the right-hand side in divergence form using the identity:

$$\nabla \cdot (\rho \nabla \log Z) = (\nabla \rho) \cdot (\nabla \log Z) + \rho \nabla^2 \log Z.$$

The drift term $-2D_\rho \gamma \nabla \rho \cdot \nabla \log Z$ can be absorbed by writing the whole equation as a divergence. Note:

$$\nabla \cdot (D_\rho \nabla \rho - v \rho) = D_\rho \nabla^2 \rho - \nabla \cdot (v \rho),$$

where $v = 2D_\rho \gamma \nabla \log Z$ is the drift velocity.

6.5 Identifying the drift velocity

From the first moment of the biased random walk:

$$v \Delta t = \int \Delta x \pi(x + \Delta x | x) d(\Delta x) = \int \Delta x p(\Delta x) (1 + \gamma \nabla \log Z \cdot \Delta x) d(\Delta x).$$

The first integral vanishes (odd moment of symmetric p); the second gives:

$$v \Delta t = \gamma \nabla \log Z \int (\Delta x \otimes \Delta x) p(\Delta x) d(\Delta x) = \gamma \nabla \log Z \cdot \sigma^2 \mathbf{I}.$$

Thus:

$$v = \frac{\sigma^2 \gamma}{\Delta t} \nabla \log Z = 2D_\rho \gamma \nabla \log Z,$$

confirming our identification.

7 Conclusion: The Keller–Segel System

7.1 The density equation

With drift $v = 2D_\rho \gamma \nabla \log Z$ we write the density evolution as:

$$\frac{\partial \rho}{\partial t} = \nabla \cdot (D_\rho \nabla \rho - \rho v) = \nabla \cdot (D_\rho \nabla \rho - 2D_\rho \gamma \rho \nabla \log Z).$$

Defining the **chemotactic sensitivity**

$$\chi := 2D_\rho \gamma, \tag{11}$$

we obtain:

$$\boxed{\frac{\partial \rho}{\partial t} = \nabla \cdot (D_\rho \nabla \rho - \chi \rho \nabla \log Z).} \tag{12}$$

7.2 The full Keller–Segel system

Together with the cue equation (8), we have derived the **Keller–Segel model**:

$$\frac{\partial \rho}{\partial t} = \nabla \cdot (D_\rho \nabla \rho - \chi \rho \nabla \log Z), \quad (13)$$

$$\frac{\partial Z}{\partial t} = D_Z \nabla^2 Z + k_p \rho r(x) - k_d Z. \quad (14)$$

Intuition

Physical interpretation of equation (13):

- $D_\rho \nabla \rho$: ordinary diffusion — cells spread out randomly.
- $\chi \rho \nabla \log Z$: chemotactic drift — cells move up the gradient of $\log Z$, i.e. toward higher cue concentrations (and hence higher value).

The competition between diffusion (spreading) and chemotaxis (aggregation) can lead to the formation of cell aggregates, a hallmark of the KS model.

7.3 Summary of the derivation roadmap

RL objective $J[\pi]$	→	Entropy-regularised optimisation
Bellman equation for V^*	→	Optimal policy π^* (softmax)
Weber–Fechner encoding $V = \frac{1}{\beta} \log Z$	→	Cue dynamics (TD update)
Taylor expansion (small steps)	→	Cue PDE: $\partial_t Z = D \nabla^2 Z + k_p \rho r - k_d Z$
Kramers–Moyal expansion	→	Density PDE: $\partial_t \rho = \nabla \cdot (D \nabla \rho - \chi \rho \nabla \log Z)$

The Keller–Segel system thus emerges naturally as the *continuum limit* of an entropy-regularised MDP solved by a population of cells performing temporal-difference learning.